CAPSTONE PROJECT REPORT

VoiceOps 2026

ZentroMart voice support agent | Implementation and evaluation summary | 25 October 2026

A customer support agent that answers from supplied documents, checks synthetic orders only after caller verification, remembers conversation context, and listens and speaks through Gnani.

What was built

The solution combines a local Ollama chat and embedding model, retrieval over ZentroMart policy and catalog documents, the provided offline order and ticket tools, and a browser microphone interface. Text and voice turns use the same conversation agent.

Step	Implementation
1 - Prompting	Local model with the ZentroMart assistant role, answer limits, grounded response instructions and structured output.
2 - RAG	Knowledge sections and catalog rows are indexed and retrieved; answers include evidence IDs and respect policy precedence.
3 - Conversation	Stateful follow-ups, clarification, phone/pincode verification, order lookup and ticket handoff via supplied mock tools.
4 - Voice	Gnani speech-to-text and text-to-speech, explicit timbre-v2.5 / Kaveri, web microphone interface and batch audio command.

Dataset scope

Validated input: 9 CSVs, 21 knowledge-base Markdown files, 52 practice questions, 23 conversations / 96 customer turns, 1,000 orders and 1,783 order items. Dataset validation passed for schema, keys, order math, practice evidence, verification gates and consistent company naming. The dataset is not packaged.

01 / BRIEF ALIGNMENT

Requirements covered

Brief requirement	Implementation and evidence
LLM baseline	Prompt-only practice mode is available for Step 1 comparison.
Document-grounded answers	RAG retrieves policy/catalog sections, preserves source IDs, prioritizes current policy and declines unsupported claims.
Conversation handling	Conversation history supports follow-ups; one clear question is asked when needed; expected policy decisions are represented.
Order and ticket tools	The supplied lookup_order and create_ticket functions are used. Sensitive order data is gated on verification; handoff is confirmed only for created=True.
Voice	Gnani is used for STT and TTS. The interface records one turn at a time; batch mode saves transcripts, text replies, source IDs and audio.
Reproducible delivery	README, requirements, .env.example and folder batch command are supplied. Dataset and credentials are excluded.

Rules and safeguards

Gnani is the only speech service; no STT/TTS backup service is used. Only the provided synthetic records are used. GNANI_API_KEY belongs in the environment, never in code or Git. Each customer clip must be no longer than 60 seconds. A user may record their own voice for tests; another person requires written consent. Test answers must come from the running agent, not be typed or edited by hand.

Caller verification

An order lookup does not disclose order details until the registered phone suffix or delivery pincode is verified. The verification value is redacted from model history and is not read back. One retry is allowed. Tools are synthetic educational simulations, not production authentication.

02 / EVALUATION

Measured quality

Practice results combine the completed qwen2.5:3b baseline with a post-fix replay over its saved retrieval contexts. Prompt-only answers were unchanged by the evidence resolver; RAG rows apply the updated policy rules to the same retrieved evidence and retain prior model fallbacks. This replay is not a fresh full model run.

Mode	Decision match	All facts	Evidence hit
Prompt-only baseline	9/52 (17.3%)	8/52	0/47
RAG post-fix replay	52/52 (100.0%)	52/52	47/47

Post-fix saved-context replay: 52/52 RAG rows contained every required fact, 52/52 matched an accepted decision, and the saved retrievals include gold evidence for 47/47 applicable questions. These counts measure the patched deterministic answer layer plus retained model fallbacks, not a full fresh inference run.

Post-fix conversation replay over the saved 96-turn suite: 96/96 all-required-fact checks; 96/96 decisions matched; 96/96 expected tool checks; 0 forbidden-content hits. The deterministic answer layer changed 2 turns; all other model/tool outputs were retained from the prior full run. The focused regression report covers 28 turns.

Defects found and fixed

The baseline exposed incomplete warranty/return answers, missed refund deadlines, and unnecessary clarification. The patched layer now answers these policy cases from retrieved sections and catalog fields, including dated policy exceptions. Replays verify the changed rules against saved retrievals; a full fresh model rerun remains outstanding, so do not describe these replays as a new end-to-end model evaluation.

A full pre-fix conversation run passed its stored checks. The post-fix replay preserved unchanged turns and rechecked turns affected by the resolver. Keep both the original evaluation JSON and post-fix replay JSON with the submission.

03 / RUN AND SUBMISSION

Operation and remaining items

After extracting the supplied dataset beside the project, install requirements and set GNANI_API_KEY as a Windows user environment variable. Do not include the dataset in the ZIP.

Batch run command

python run.py --audio-folder --language en-IN --output sample_outputs

The command accepts WAV, MP3, M4A, FLAC or OGG and saves each transcript, reply text, evidence IDs and Gnani audio under sample_outputs//. The browser microphone interface starts with python run.py --web --language en-IN.

Limits and next improvements

The local model is variable and sometimes omits required facts despite retrieving the right evidence. The solution uses mock order/ticket tools and has no live inventory or production identity checks. Next: improve retrieval/answer validation and test more intents across repeated runs. Only synthetic records are used.

Deliverable	Status
Code, run command, README, requirements, .env.example	Prepared in local package; no dataset or secrets included.
Evaluation evidence	Post-fix saved-context replays: practice 52/52 facts and 52/52 decisions; conversation 96/96 facts, 96/96 decisions and 96/96 tool checks. Fresh full model reruns remain unverified.
Report, maximum 4 pages	Updated with baseline results, post-fix replays and their limitations.
Demo video, maximum 2 minutes	Still requires a screen recording: grounded answer, follow-up, and refusal or human handoff. The submitter need not appear on camera.
Public Git repository URL	Must be published and added to the README before submission.

The implementation, test outputs and report are packaged as a draft. Final submission still requires demo.mp4 and the public GitHub URL in the README. The saved-context replays are not a substitute for a fresh full model run; disclose that limitation.

Source: VoiceOps 2026 Capstone Brief, pages 3-14; supplied synthetic dataset and validator.